

# Assignment-3.4

## “Component- specific troubleshooting”

**TASK 1.** Open your desktop or laptop, locate the RAM slots, and remove and re-insert the RAM module to simulate reseating — then document each step you took and any issues you observed during the process.

1. **Safety and Preparation:**

Shut down the computer completely, turn off the power supply switch at the back, and unplug the main AC power cable. Press and hold the power button for 10 seconds to drain any leftover electrical charge inside the power supply.

2. **Opening the Case:**

Remove the side panel of the computer case to expose the motherboard. Touch an unpainted metal part of the PC frame to ground yourself and prevent static discharge.

3. **Locating and Removing the RAM:**

Find the rectangular memory sticks seated near the CPU socket. Gently push down on the plastic retention clips at both ends of the RAM slot until they snap open. Carefully lift the RAM module straight up out of the slot by holding its outer edges.

4. **Re-inserting (Reseating) the RAM:**

Align the notch on the bottom edge of the RAM stick with the raised key inside the motherboard slot. Insert the module evenly into the slot and press firmly downward on both ends until the side clips click back into place automatically.

5. **Testing System Boot:**

Close the case, reconnect the power cable, turn on the PSU switch, and press the power button.

**Initial Observation:** If the RAM is not pushed in hard enough, the side clips will not fully close, and the PC will turn on with fans spinning but display a black screen (no POST display) alongside motherboard warning beeps or memory diagnostic LEDs.

**Final Result:** Once firmly snapped into place, the computer successfully powers on, displays the motherboard logo screen, and loads the operating system normally

**TASK 2.** Create a troubleshooting checklist for diagnosing a faulty GPU in a gaming PC, including at least 5 steps from initial symptom identification to confirming the fix. Think about display issues, driver checks, and physical connections.

1. **Identify the Physical Symptoms:**

Check if the monitor displays "No Signal," shows strange colored flickering lines/dots (visual artifacts), or if the screen goes black specifically while launching 3D games.

2. **Check External Display Connections:**

Verify that the DisplayPort or HDMI cable is plugged directly into the dedicated GPU ports on the back of the case (and not into the motherboard's video port). Try swapping to a known good display cable or testing with a different monitor.

3. **Inspect Internal Hardware & Power Connections:**

Power off the PC and inspect the GPU inside the case. Ensure the PCIe power cables (6-pin / 8-pin / 16-pin) are fully clicked into the card. Unclip and reseal the graphics card in the PCIe slot to ensure a solid connection. Check if the GPU cooling fans spin up when powered on.

4. **Clean and Update Display Drivers:**

Boot Windows into Safe Mode and run **Display Driver Uninstaller (DDU)** to completely wipe old driver files. Restart normally and perform a clean installation of the official graphics driver from NVIDIA, AMD, or Intel.

5. **Confirm the Fix with Stress Testing:**

Run a GPU stress test or launch a demanding 3D gaming benchmark (such as 3DMark or FurMark) for 15–20 minutes. Monitor temperatures and ensure there are no crashes, visual artifacts, or black screen errors.

**TASK 3.** Simulate a storage issue by unplugging and replugging a SATA or NVMe drive (SSD/HDD) on a test system, then describe the error messages or BIOS indicators you see before and after reconnecting.

### Observations Before Reconnecting (Drive Unplugged):

- **SATA / NVMe Disconnected State:** The physical hard drive or SSD is completely detached from the motherboard's SATA port or M.2 NVMe slot.
- **Error Messages Shown:**
  - On power up, the system bypasses Windows and displays a black screen error reading:  
Reboot and Select proper Boot device or No Boot Device Found. Press any key to reboot.
- **BIOS Indicators:**
  - Upon entering the BIOS setup screen (**Del** or **F2**), the **System Overview** / **Storage Info** page shows **N/A**, **Not Present**, or **Empty** for that specific M.2 or SATA slot. The drive is missing completely from the Boot Priority List.

### Observations After Reconnecting (Drive Plugged Back In):

- **SATA / NVMe Reconnected State:** The drive is securely reinserted into its socket, locked down with its screw/latch, and connected to power/data cables.
- **BIOS Indicators:**
  - Re-entering the BIOS shows the drive correctly identified with its full name and capacity (for example: *Samsung SSD 980 1TB* or *WDC 500GB HDD*). The drive automatically reappears under the Boot Priority order.
- **System Status:**
  - The PC boots directly into the Windows desktop operating system without displaying any boot errors.

**TASK 4.** Given a scenario where a PC randomly restarts, list which component (RAM, GPU, storage) you would check first and explain your reasoning, then outline the specific diagnostic steps you would perform for that component.<sup>4</sup>

**Reasoning:**

While a failing GPU or faulty storage drive usually results in game crashes, display artifacts, or frozen screens with specific error codes, **RAM memory errors are the single most common cause of random, sudden system restarts** without warning. When the CPU attempts to read or write data to a corrupted memory sector, it triggers a fatal kernel error (BSOD) or forces an instant system reset to prevent data corruption.

**Step-by-Step Diagnostic Plan for RAM:**

1. **Physical Inspection and Reseating:**  
Turn off the PC, disconnect power, remove the memory modules, clean any dust out of the slots, and firmly reinstall them into the motherboard's recommended slots (usually slots 2 and 4 for dual-channel setups).
2. **Single-Stick Isolation Test:**  
If the computer has multiple RAM sticks installed, remove all sticks except one. Run the PC to see if the random restarts continue. Swap sticks one by one to isolate a single defective RAM module.
3. **Run Hardware Diagnostic Software (MemTest86):**  
Create a bootable USB drive using **MemTest86**. Boot the PC directly from the USB drive before Windows loads and execute a full pass test.
4. **Evaluate Diagnostics Result:**  
If MemTest86 reports even a single memory error, the RAM stick is defective and must be replaced. If zero errors are found after multiple passes, proceed to test secondary suspects like CPU temperatures or the Power Supply Unit (PSU).